Coin Grid

Time limit: 1.00 s **Memory limit:** 512 MB

There is an $n \times n$ grid whose each square is empty or has a coin. On each move, you can remove all coins in a row or column.

What is the minimum number of moves after which the grid is empty?

Input

The first input line has an integer n: the size of the grid. The rows and columns are numbered $1, 2, \ldots, n$.

After this, there are n lines describing the grid. Each line has n characters: each character is either `.` (empty) or `o` (coin).

Output

First print an integer k: the minimum number of moves. After this, print k lines describing the moves.

On each line, first print 1 (row) or 2 (column), and then the number of a row or column. You can print any valid solution.

Constraints

- $1 \leq n \leq 100$

Example

Input	Output
3 ..o o.o ...	2 1 2 2 3